



A new single-ion polymer electrolyte based on polyvinyl alcohol for lithium ion batteries

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ABSTRACT

In this study, a novel single-ion conducting polymer electrolyte, lithium polyvinyl alcohol oxalate borate (LiPVAOB), from the reaction of poly(vinyl alcohol) (PVA) with different molar ratio of boric acid, oxalic acid and lithium carbonate was prepared. The prepared materials were characterized by Fourier transformation infrared spectroscopy (FT-IR), X-ray diffraction (XRD), scanning electron microscopy (SEM), thermogravimetry (TG), differential thermal analysis (DTA), electrochemical impedance spectroscopy (EIS) and linear sweep voltammetry. Ionic conductivity of these polymer electrolytes from adding an assistant (about 20 wt.% propylene carbonate) is dependent on molar ratio of the reactants and can up to $6.11 \times 10^{-6} \text{ S cm}^{-1}$ at ambient temperature. Their electrochemical window can be stable up to 7 V (vs. Li^+/Li), which is of great attraction for high voltage lithium ion batteries with high energy density.

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1. Introduction

Lithium ion batteries have gained an unprecedented importance in the last several decades as the energy storage of portable devices such as cell phones and laptop computers [1,2]. Furthermore, lithium ion batteries are also considered as the optimal power sources for hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEVs) and electric vehicles (EVs) [3,4]. However, most lithium ion batteries comprising of liquid electrolytes have potential safety problems due to the existence of large amount of organic solvents. In order to solve the safety problems, polymer electrolytes (PEs) have received great attention. In the case of application of solid PEs, there is still quite some space because of the low ionic conductivity [5]. Gel PEs, which have the characteristics of the solid and liquid electrolytes, have got increasing attention for its good ionic conductivity, wide electrochemical window, good compatibility with the electrodes, good thermal stability and so on [6–8]. Poly(vinylidene fluoride) (PVDF) [9–11] and its copolymer poly(vinylidene fluoride-co-hexafluoropropylene) (P(VDF-co-HFP)) [12–15], poly(ethyleneoxide) (PEO) [16,17], polyacrylonitrile (PAN) [18], poly(methyl methacrylate) (PMMA) [19] have been widely studied as polymer matrices for the preparation of gel PEs in order to make them feasible for commercial applications.

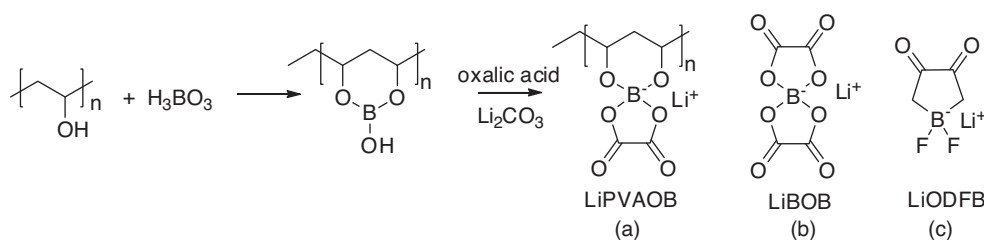
The mechanism of ion conduction in solid PEs was explained based on dissolving alkali metal salts in polyether matrix [20,21]. Ion conduction takes place in the amorphous phase of polymers such as polyethylene oxide and ether oxygen atoms act as ion-coordinating sites. The ether oxygen atoms interact with the cations (Lewis acid–base interactions) and cause salt solvation. The cation transport is assisted by segmental motion of the polymer chains. The counter anions in the polymer electrolytes have weak interactions with the polyethers, hence transport more easily compared to cations. Thus a major drawback of dual ion conducting in PEs is the low lithium ion transference number (0.3–0.5) [21]. In the same time, the mobility of anions results in polarization and reduces the lithium ion transference number. Therefore it is necessary to design single-ion conducting polymer electrolytes to guarantee the stability of anions to gain good performance for lithium ion batteries.

To date, two approaches have been reported to reduce the mobility of anions. The first one deals with introduction of interacting sites that preferentially interact with the anions [22,23]. The other is to anchor anions to the polymer backbone [24,25], which is a common method to achieve single-ion conducting PEs. However, it should be noted that the ionic conductivity of single-ion PEs is generally lower than that of dual ions conducting system because of the lost conductivity contribution from anion transference. In order to improve their ionic conductivity, various methods have been tried [22,26,27]. However, the subject still remains as a big challenge and there have been very few publications on single-ion polyester-based electrolytes.

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Scheme 1. (a) The preparation process of LiPVAOB and the structures of (b) LiBOB and (c) LiODFB.

Polyvinyl alcohol (PVA) is a water-soluble polyhydroxyl polymer whose physical properties are dependent on the degrees of polymerization and alcoholysis. PVA has an excellent chemical resistance, thus it can be used in many practical applications such as adhesives, textiles, and pharmaceutical and biomedical industries due to nontoxic and biodegradable properties [28,29]. It can also be modified by chemical reactions, which is a highly versatile method to improve the chemical, thermal, and mechanical properties [30]. One of the most common methods is the reaction of hydroxyl groups of PVA with aldehydes to form acetal or hemiacetal [31]. PVA can also physically cross-link with boric acid through hydrogen bonding and ionic interactions in the hydrated state [29,32].

Lithium chelatoborates as electrolyte salts have been widely studied and its aim is to substitute LiPF_6 [33–36]. The latter is unstable at elevated temperature and sensitive with trace amount water [37]. Among the borates, lithium bis(oxalate)borate (LiBOB) and lithium oxalate difluoroborate (LiODFB) (Scheme 1) are the promising candidates used as electrolyte salts in lithium ion batteries [36,38]. The merits of these lithium salts are not only the low production cost, wide potential window, good thermal stability but also they can form effective solid electrolyte interface (SEI) to stabilize the graphitic anode structure. LiBOB and LiODFB have the similar structure based on an oxalate-chelated borate and present good electrochemical performance. However, little attention has been paid on lithium chelatoborates polymer electrolytes, let alone on the single-ion conductor. From this point of view, lithium chelatoborate-based polymer electrolytes may be an alternative system to promote the applications of lithium rechargeable battery.

In this work, we reported the synthesis and electrochemical properties of a novel single-ion conducting polymer electrolyte, lithium polyvinyl alcohol oxalate borate (LiPVAOB, Scheme 1a), whose monomer is structurally similar to LiBOB (Scheme 1b), based on PVA, boric acid (H_3BO_3), lithium hydroxide (LiOH) and oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$). The structure, thermal property, morphology and electrochemical property of the as-prepared single-ion conducting PEs have been investigated and the primary results show that the prepared PEs have good electrochemical performance.

2. Experimental

The polymer membrane was synthesized by the reaction of PVA (degree of alcoholysis $\geq 98\%$, Mw: 105,000), lithium carbonate, boric acid, oxalic acid according to the chemical equation (Scheme 1a). All chemicals were provided from Sinopharm Chemical Reagent Co., Ltd. Boric acid and PVA were dissolved into DMSO and heated to 80°C for 5 h to give a homogeneous and transparent solution. Then lithium carbonate and oxalic acid were added. The mixture was stirred at 100°C for 24 h. Cooled to room temperature, the solution was later cast on a flat glass plate at 70°C to remove the solvent to get membranes with a thickness of $50\text{--}70\ \mu\text{m}$. The membranes were punched into circular pieces ($d = 19\text{ mm}$) and further dried under vacuum at 70°C for 48 h to remove the trace amount of solvent. Later, the pieces were soaked in propylene carbonate

(PC) over 12 h in a glove box (water content $< 1\text{ ppm}$) to get the PEs for further measurement. LiPVAOB based on different molar ratios of reactants ($n\text{-OH}/n\text{H}_3\text{BO}_3/n\text{Li}_2\text{CO}_3/n\text{H}_2\text{C}_2\text{O}_4 = 2/0.9/0.45/0.9, 2/0.95/0.475/0.95, 2/1/0.5/1, 2/1.05/0.525/1.05, 2/1.1/0.55/1.1$) were marked as LiPVAOB_{0.9}, LiPVAOB_{0.95}, LiPVAOB_{1.0}, LiPVAOB_{1.05} and LiPVAOB_{1.1}.

The conventional KBr disk method was employed to measure the Fourier transformation infrared spectroscopy (FT-IR) of the membranes. Before FT-IR spectra measurements, the samples were dried at 70°C under vacuum for 12 h and then stored in the dry box for 2 h. The IR spectra were recorded with a BRUKER VECTOR-22 spectrometer. X-ray diffraction (XRD) was performed on a Bruker D8 X-ray diffractometer (50 kV, 50 mA) with a copper target ($\lambda = 1.5405\text{ \AA}$) and Ni filter at a scan rate of 5° min^{-1} from 10° to 90° . Thermogravimetric analysis (TGA) and differential thermal analysis (DTA) of the polymers were carried out by utilizing a Perkin-Elmer TGA7/DTA7. The samples were heated from room temperature to 400°C at a heating rate of $10^\circ\text{C min}^{-1}$ under nitrogen atmosphere. The surface morphology of the prepared membranes was investigated by Philips XL30 scanning electron microscopy (SEM). The membranes were dipped into liquid nitrogen and broken into two parts, and then the SEM micrographs of the cross-sections were taken. All of the samples were sputtered with gold prior to the SEM measurement. The thickness of the membranes was measured with a micrometer (SM & CTW, Shanghai).

The ionic conductivities of the PEs were measured by the electrochemical impedance spectroscopy (EIS) method. The samples were prepared in the glove box under argon atmosphere with a humidity $< 1\text{ ppm}$. They were measured in blocking-type cells which were fabricated by sandwiching the membranes between two stainless steel electrodes. The experiments were performed in a cylindrical cell of an electrode diameter of 1.80 cm. Impedance data were obtained with a computer interfaced with an electrochemical working station CHI660C (Chenhua) in the frequency range from 10 Hz to 100 kHz. The cells were placed in a temperature-controlled oven and the impedance responses were recorded in the range of $25\text{--}85^\circ\text{C}$. The linear sweep voltammetry for the as-prepared PEs was carried out on a two-electrode cell. Stainless steel was used as the working electrodes and lithium foil was used as the counter and reference electrode, respectively. The measurement was done between 0 and 7 V (vs. Li^+/Li) at a scan rate of 2 mV s^{-1} at 25°C with an electrochemical working station CHI660C (Chenhua).

3. Results and discussion

The reaction of PVA with H_3BO_3 , Li_2CO_3 and $\text{H}_2\text{C}_2\text{O}_4$ is expected to take place well under a mild condition as shown in Scheme 1. It is necessary to control the reaction conditions to ensure the obtaining of the product LiPVAOB and the addition serial of the reactants is very important. PVA is a water-soluble polymer and cannot dissolve well in some organic solvents. However, the amount of water is strictly controlled in lithium batteries. To get the target material, the polymer must react with organic materials, inorganic compounds and salts. As a result, DMSO, an aprotic polar solvent which

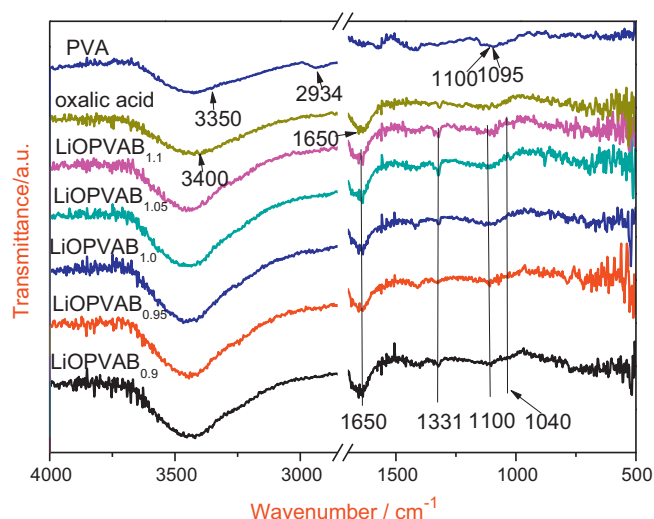


Fig. 1. FT-IR spectra of PVA, oxalic acid and LiPVAOB.

can dissolve polar and nonpolar compounds even inorganic salt, was used as the reaction media in the procedure. The concentration of PVA in the DMSO solvent is low, 1 wt.%. The reaction activities of PVA and oxalic acid were very different when they reacted with boric acid. Hydroxyl group in PVA reacted with boric acid to give borate were a little more difficult than oxalic acid with boric acid to give LiBOB. So the reaction was controlled by the order of adding chemicals. At the beginning, PVA reacted with boric acid for about 5 h at 80 °C to give the intermediate borate. This process, of course, is not difficult since the hydroxyl groups in poly(ethylene glycol) can react with boric acid to get polymer electrolytes containing borate ester groups [39]. Then the borate reacted with lithium salt to get the target polymers.

Fig. 1 shows the FT-IR spectra of PVA, oxalic acid, and the LiPVAOB membranes. Characteristic absorptions of the virginal PVA are located at 3350, 2934, and 1100 cm^{-1} , corresponding to ν_{OH} , $\nu_{\text{C-H}}$, and $\nu_{\text{C-C-O}}$, respectively. The peak at 1095 cm^{-1} can be attributed to C–O stretching [32]. Characteristic absorptions of the pure oxalic acid are located 3400 and 1650 cm^{-1} corresponding to ν_{OH} and $\nu_{\text{C=O}}$. In all spectra of LiPVAOB, an absorption peak appeared at 1650 cm^{-1} was attributed to $\nu_{\text{C=O}}$. The absorption bands at about 1040 and 1331 cm^{-1} were assigned to the vibration of the B–O–C bond and the anti-symmetric stretching vibration of the B–O bond, respectively [40]. The absorption peak at about 1100 cm^{-1} belongs to $\nu_{\text{C-C-O}}$ in the polymer membranes. These characteristic peaks indicate that the preparation of LiPVAOB is successful. The peak of 3400 cm^{-1} is the ν_{OH} caused by water absorbed by LiPVAOB in air due to the existence of Li^+ ions and its strong polarity.

The X-ray diffraction (XRD) patterns of the virginal PVA and the prepared LiPVAOB are shown in Fig. 2. It is well known that the virginal PVA exhibits a semicrystalline structure due to the high hydrogen bond of the hydroxyl groups. Three peaks were found in the XRD pattern of PVA. One peak with high intensity appears at about $2\theta = 19.5^\circ$ and the other peaks with low intensity appear at about 23° and 41° , which correspond mainly to [1 0 1], [2 0 0] and [1 1 1] planes [41,42]. After the reaction to prepare the polymer LiPVAOB, the peak intensity at $2\theta = 19.5^\circ$ (for [1 0 1] plane) is reduced, which indicates that the crystallinity of PVA matrix is decreased since the distance between the C–C chain is enlarged [42]. It can also be seen from Fig. 2 that the peak at 41° for PVA disappeared completely after the reaction of the hydroxyl groups with boric acid. Because the hydroxyl groups of PVA reacted with boric acid and boric acid chelated lithium oxalate to give the structure just

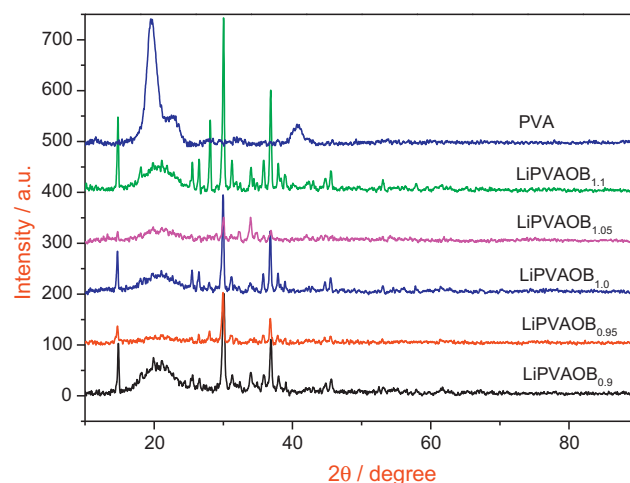


Fig. 2. X-ray diffraction patterns of PVA and LiPVAOB.

like LiBOB in the long carbon chain, the polymer membranes exist in a semicrystalline structure, and some new peaks at about 18° , 30° and 37° appear. As to the specific diffraction peaks, further theoretical simulation is needed.

The thermal stability is an important property for polymers electrolytes during application in lithium ion batteries. As seen from Figs. 3 and 4, two major peaks are found at about 223 and 300 °C in the DTA curve of PVA. The 223 °C peak represents the melting temperature of semicrystalline PVA polymer. The 300 °C peak is related to carbon chain decomposition reaction [43]. There are also two peaks in the DTA curves of LiPVAOB membranes. The peak at 100 °C corresponds to the glass transition of the carbon chain, which is similar to PVA, and slight decomposition of the polymers. The peak at 170 °C represents the main decomposition of the polymers since the stability of this polymer is a little less than that of PVA (about 220 °C). The results are consistent with the TG curves. The LiPVAOB membranes are thermally stable up to nearly 100 °C. In the case of lithium ion batteries, the thermal stability can be up to 85 °C by adding some fire-retardants and other additives [44]. When the temperature is high above 100 °C, the LiPVAOB begins to decompose little by little, and there is no clear stage of weight loss till 400 °C. The degradation process of the membranes will absorb much heat, which is also good for the safety of the battery systems. Additionally, the TG–DTA curves of LiPVAOB also give us a good

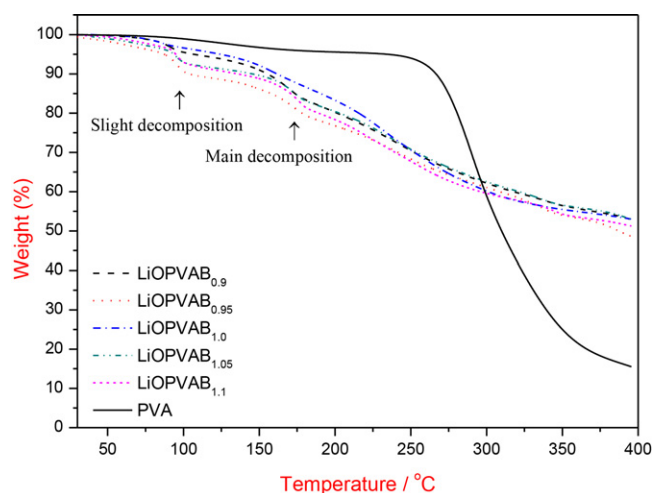


Fig. 3. Thermogravimetric analysis (TGA) profiles of PVA and LiPVAOB: rising temperature rate is $10^\circ\text{C min}^{-1}$.

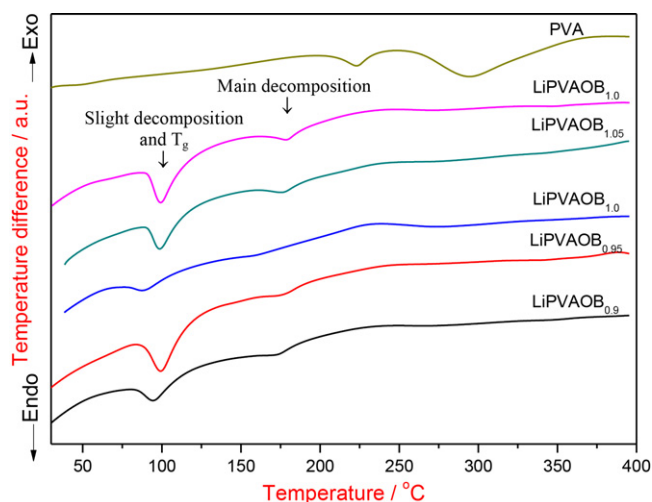


Fig. 4. Differential thermal analysis (DTA) curves of PVA and LiPVAOB: rising temperature rate is $10^{\circ}\text{C min}^{-1}$.

proof that there is no the byproduct, LiBOB, formed in the polymer membranes. LiBOB is stable to 302°C where it decomposes rather than melts [35]. The LiPVAOB begins to decompose at about 100°C and there is a wide endothermic peak in the curve of DTA. When LiBOB decomposes at about 300°C , there should be an endothermic peak in the DTA curve. However, no any peaks appear from 200 to 350°C in the DTA curve. This suggests that LiBOB does not exist in the polymer.

The morphology of the virginal PVA and LiPVAOB membranes (LiPVAOB_{1.0} as the example) was examined by SEM and the results are shown in Fig. 5. The surface of the virginal PVA is clear and homogeneous. After the reaction, the surface of the LiPVAOB_{1.0} membrane (Fig. 5b) becomes very rough and compact. However, from the cross-section (Fig. 5c), the inner structure of the membrane is uniform, which guarantees the polymer has good mechanical property. In the case of gel PEs, they are porous and

Table 1

The amount of PC uptake (η) of different membranes at 25°C .

LiPVAOB	W_0 (mg)	W_t (mg)	$W_t - W_0$ (mg)	η
0.9	24.3	30.0	5.7	23.4%
0.95	24.6	30.6	6.0	24.4%
1	23.0	29.2	6.2	26.9%
1.05	23.5	30.1	6.6	28.1%
1.1	24.1	29.7	5.6	23.2%

their mechanical strength is poor, which limits their practical application [4,5,11,12]. Adding PC of high-permittivity to the polymer membranes can increase ionic conductivity. It is known that added solvents may destroy the good morphology of the polymer membranes. From Fig. 5d, it can be seen that the morphology hardly changes after immersing in PC for 12 h. All these can prove that the LiPVAOB that we synthesized is competent to be the electrolyte in lithium ion batteries. It is known that adding high-permittivity molecules PC to PEs is one successful approach to increase the amount of the amorphous phase and ionic conductivity. However, it will bring side effects such as reduced dimensional stability, mechanical strength and poor film [45]. Our prepared PEs can overcome its shortcomings because of special chemical structure. The amount of uptake (η) is calculated as Eq. (1):

$$\eta = \frac{W_t - W_0}{W_0} \times 100\% \quad (1)$$

where W_0 and W_t are the weight of the films before and after absorbed PC. Table 1 shows the different η value of different LiPVAOB at 25°C . It can be seen that the amount of the PC uptake is not high, about 20%, higher than that for the matrix from poly(acrylic acid) [46] and much lower than those for other gel PEs [9–15,18,19]. The maximum η belongs to LiPVAOB_{1.05}, that is, the molar ratio of $-\text{OH}/-\text{B} = 2/1.05$. It should be pointed out that the amount of boric acid cannot be more excessive.

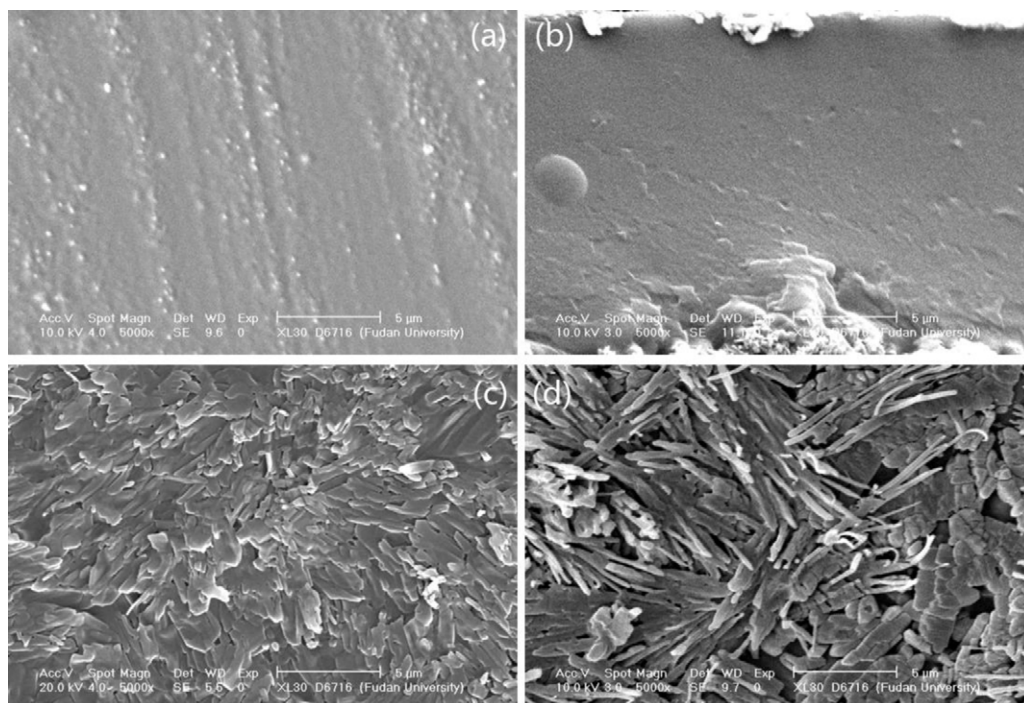


Fig. 5. SEM micrographs of the surfaces of (a) the virginal PVA and (b) LiPVAOB_{1.0}, and the cross-sections of LiPVAOB_{1.0} (c) before and (d) after immersing in PC for 12 h.

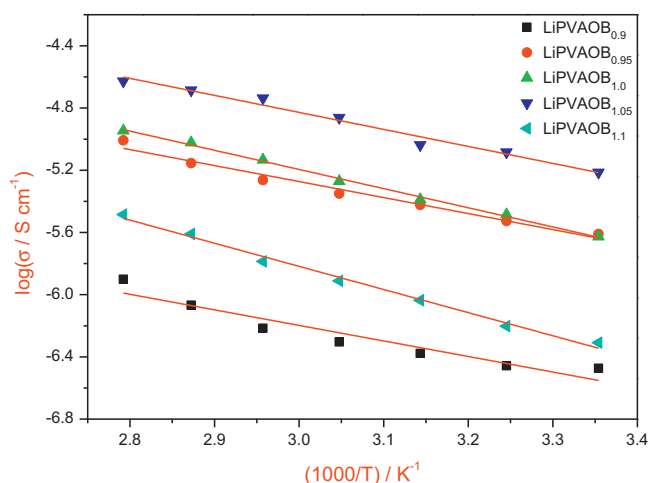


Fig. 6. Arrhenius plot of the prepared LiPVAOB polymer electrolytes.

Fig. 6 shows Arrhenius representations from 289 to 398 K for the temperature dependence of the ionic conductivity. The ionic conductivity is calculated using the following relationship:

$$\sigma = \frac{l}{R_b A} (S \text{ cm}^{-1}) \quad (2)$$

where σ is the ionic conductivity, l is the thickness of the polymer electrolyte, R_b is the resistance of the PEs which can be got by EIS and A is the surface area of the electrode. The highest ionic conductivity at 298 K is about $6.11 \times 10^{-6} S \text{ cm}^{-1}$ for LiPVAOB_{1.05} polymer electrolyte, which is almost 20 times as high as that of LiPVAOB_{0.9}. At 398 K, the ionic conductivity of LiPVAOB_{1.05} polymer electrolyte is about $3.3 \times 10^{-5} S \text{ cm}^{-1}$. The ionic conductivities of the PEs increase with the temperature, and their relationship can be expressed as Eq. (3):

$$\sigma = \sigma_0 \exp \left(\frac{-E_a}{RT} \right) \quad (3)$$

where σ_0 is the pre-exponential factor and E_a is the activation energy. The activation energy has been found to be 0.035, 0.036, 0.043, 0.039 and 0.052 eV for LiPVAOB_{0.9}, LiPVAOB_{0.95}, LiPVAOB_{1.0}, LiPVAOB_{1.05} and LiPVAOB_{1.1}, respectively. As to the action of the PC solvent, it is similar to that of water for proton exchange membrane. The PC molecular just acts as an assistant instead of a plasticizer since the polymer is not plasticized and Li^+ ions, like protons, are

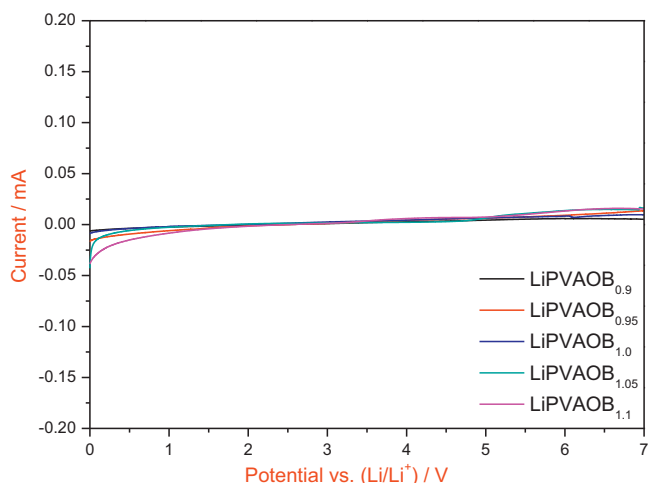


Fig. 7. Linear sweep voltammograms of the prepared LiPVAOB polymer electrolytes.

fixed at the polymer chain instead of free lithium salts in the traditional gel PEs [4,5,11,12].

As can be seen from Fig. 7, no current peak is found through the working electrode from open circuit potential to 7 V (vs. Li^+/Li) for all the PEs. It also shows that the amount of water and DMSO is very rare. This reveals that the voltage breakdown of all the as-prepared PEs, which is mainly regarded as the decomposition of electrolytes owing to the oxidation of the anions, is 7 V (vs. Li^+/Li). That is, the electrochemical window of LiPVAOB polymer electrolytes can be up to 7 V (vs. Li^+/Li), which is larger than the borate ester, 5.8 V [23], and can meet the requirement for application in lithium polymer battery of high voltage.

4. Conclusions

We have developed a novel single-ion conducting polymer electrolyte membrane, LiPVAOB, on the basis of different molar ratio of polyvinyl alcohol with boric acid, oxalic acid and lithium carbonate. The obtained membranes show good thermal properties and good morphology even after uptaking some amount of propylene carbonate (PC). When the molar ratio of $-\text{OH}/-\text{B}$ is 2.0/1.05, the given product, LiPVAOB_{1.05}, shows high ionic conductivity up to $6.11 \times 10^{-6} S \text{ cm}^{-1}$ at 298 K and stable electrochemical window up to 7 V (vs. Li^+/Li). The polymer electrolytes with higher Li^+ content can meet Arrhenius behavior, indicating that the Li^+ ion motion is depended on the flexibility of the polymer side chain. These materials have high ion conductivity and can be suggested for application to lithium ion batteries.

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